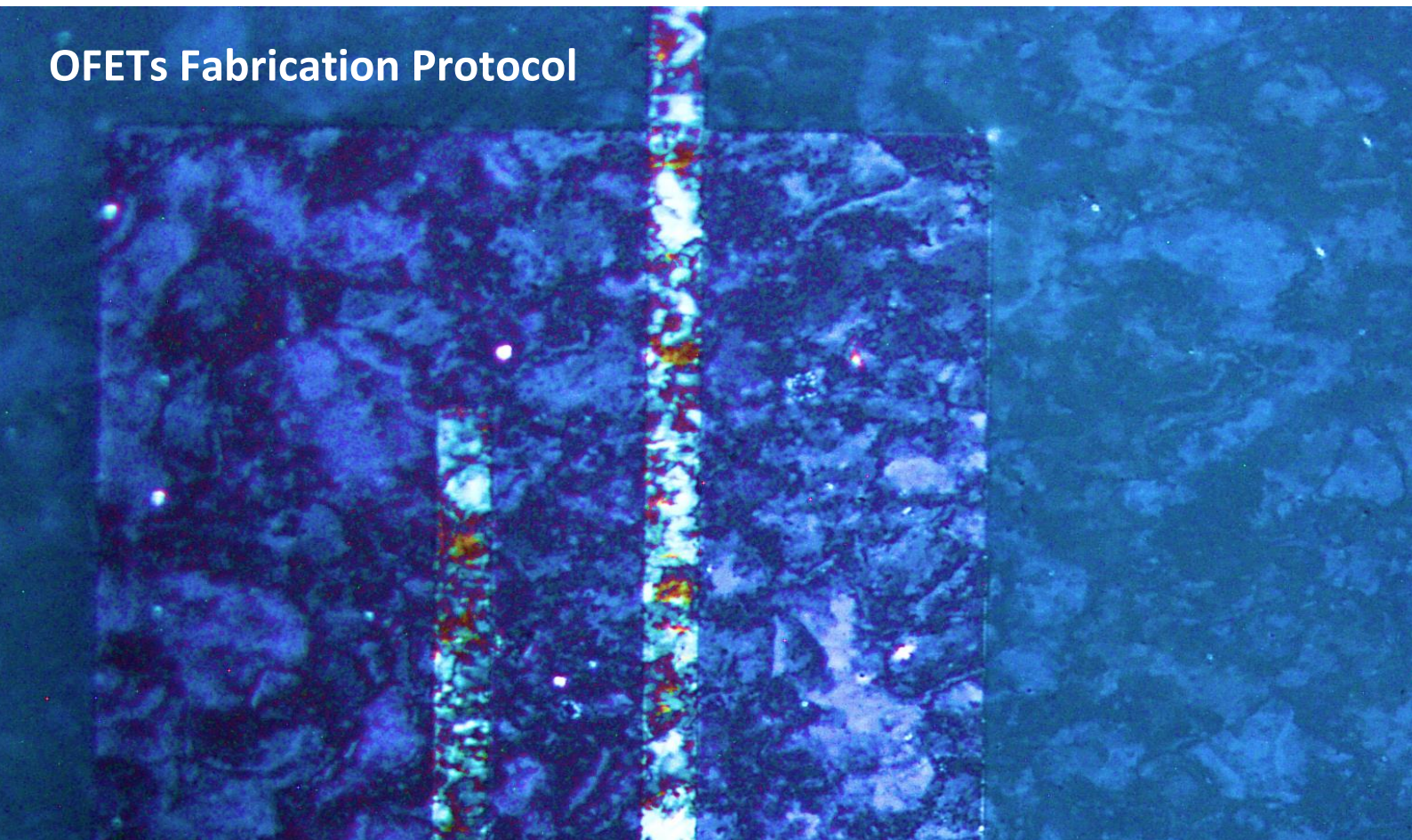


OFETs Fabrication Protocol



January 2022

General consideration: Avoid leaving the wafer in air. For that, leave the wafer immersed in ISOPROPANOL.

1. PHOTOLITHOGRAPHY

a. **Wafer cleaning.** We clean the wafer as follow using:

- Rinsing with Acetone
- Rinsing with Isopropanol
- Drying with Nitrogen gun

b. **Resist.**

- Clean the coater and the support with Acetone
- The support must be fixed with the angle screwdriver. Besides, a **square of double-tape** is put on it to be sure that the wafer won't move.
- The coating parameters are: 4000rpm/s (P15 must be on **0.7**), 5000rpm (P16 must be on **83.30%**), 25 seconds.
- The wafer is positioned on the **center** of the support and vacuum need to be created again between the support and the wafer in order to fix the wafer, it is maintained for the whole process.
- The resin (SHIPLEY S1813) is manually deposited on the wafer's surface and must cover it completely (attention to bubbles! Recommendation: pour the resin directly without the help of the pipette). After the deposition of the resin, we must immediately activate the rotation pressing START. DO NOT press manually the lid of the spinner. Once the deposition process is done, remove the wafer very carefully, it is very fragile. NOTE: if the deposition process goes wrongly, the wafer can be reused again by cleaning with acetone (take care of scratches)
- Anneal the resin by putting the wafer on the hot plate at 75 °C for 1 min.

c. **Laser Writing**

- See notes on how to use the software. We recommend the following 9 focus points: 0,0 30,0 -30,0 0,30 0,-30 20,20 20,-20 -20,-20 -20,20

d. **Developing**

- Reveal the pattern on the resin and dissolve all the resin that has not been exposed to the laser. To do this, we immerse the wafer within a beaker with the "REVELADOR" for the SHIPLEY S1813 and stir it manually for 40s-60s.
- We then clean the wafer by **thoroughly** rinsing with distilled water and drying with the Nitrogen gun. (NOTE: dry thoroughly the **front and back** of the wafer. Remaining water may delay/affect the time need during the evaporation).
- Store the wafer in the transparent box completely covered with a foil of Aluminum. Write on the top of box your name and the parameters to use for the evaporation of the electrodes. (Note: 5 nm Cr + 40 nm Au). In order to speed up the process, please call/inform the technician once the wafer is ready for the evaporation.

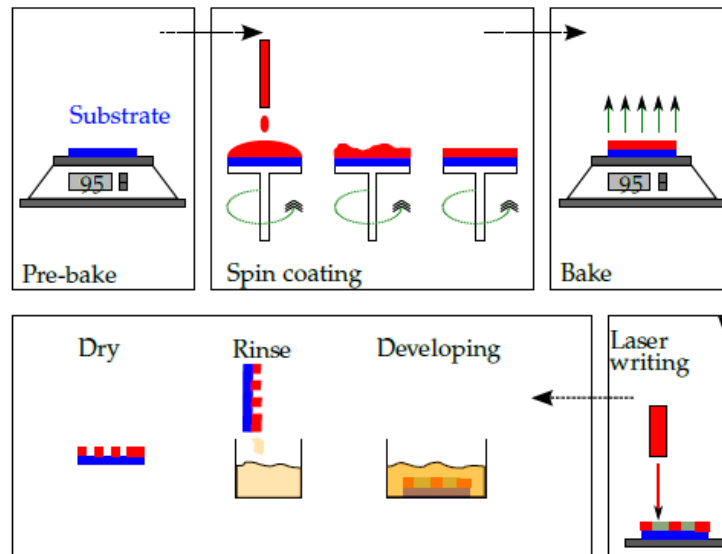


Figure 3.1: Lithography experimental procedure scheme.

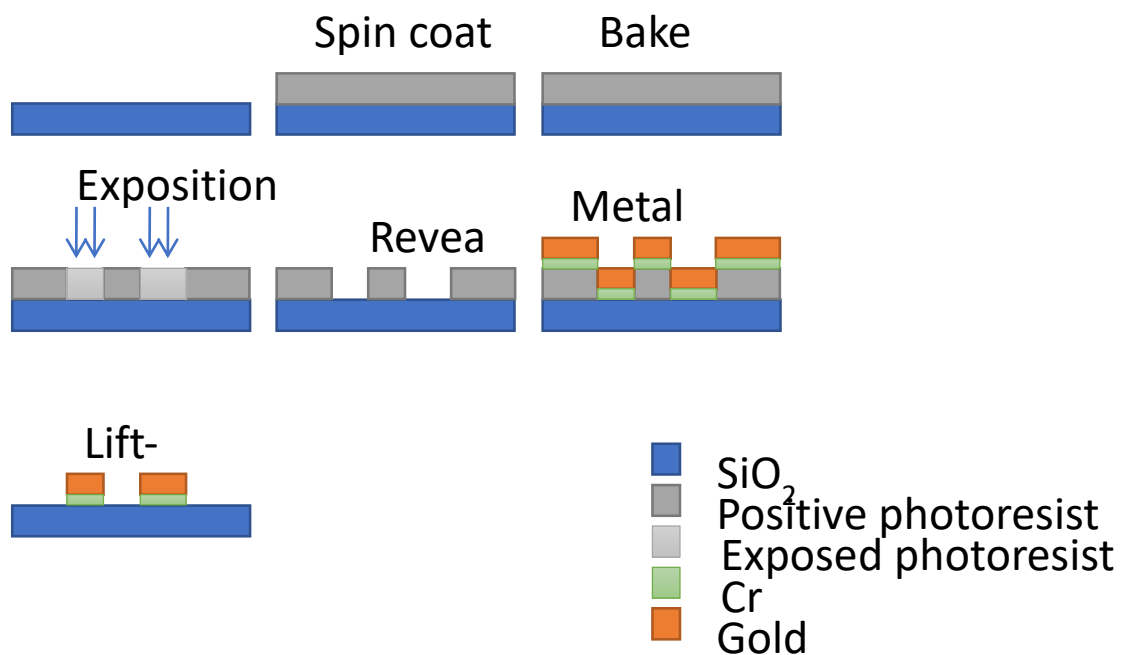
2. Cr/Au EVAPORATION – Done by the clean room technicians

- We usually use 5nm Cr + 40nm Au. NOTE: if the evaporated gold has a pinkish tone, please inform the technician.

3. Once the evaporation is done, CUT THE WAFER to create SMALLER SAMPLES. Try to do not scratch the samples while cutting. (Note: Use a wipe to protect the wafer from silicon dust/ruler...)

4. LIFT-OFF. Right after the Cr/Au Evaporation

- 3 baths of ACETONE in ultrasound, each one for 15 minutes or 1 bath in ACETONE for 20 minutes (use the **cotton** stick in acetone to get rid of extra gold just in case of emergency). For each bath, use fresh acetone.
- 3 baths of ISOPROPANOL in ultrasound, each one for 15 minutes or 1 bath in ISOPROPANOL for 20 minutes. For each bath, use fresh isopropanol.
- Leave the wafer in ISOPROPANOL until it is used (if you are going to use them the same day or the next). Otherwise, store them in the glovebox and clean again right before SAM following the same procedure)



5. PREPARATION OF THE SAM

- For each sample, prepare the vials for the SAM. Clean the vials with ISOPROPANOL by rinsing and prepare solutions of PFBT and ISOPROPANOL. The concentration must be $15 \times 10^{-3} M$ (i.e. $2 \mu\text{L/mL}$). Then, in each vial, we will have 20mL of ISOPROPANOL and 40 μL of PFBT. (Alternative: it is possible introduce in one vial two samples max. Be careful about scratching the gold electrodes, put them back to back)
- Waste: throw away the solutions in NO HALOGENATED.

6. GOLD-ACTIVATION and MODIFICATION

- Put the samples inside the UVO cleaner (Ozone treatment) to activate the surface for 25 minutes. Do not put the plastic box inside, rather an aluminum foil containing all the samples.
- Immediately** after the Ozone treatment, immerse the samples within the SAM solutions. Leave them for 15 minutes (one sample in each vial). It is **mandatory** to bring the SAM vials next to the UVO cleaner instead of bringing the samples to the lab.
- Rinse the samples with fresh ISOPROPANOL and leave them in IPA until the coating. (Recommendation: **DO NOT LEAVE** the samples for a long time immersed in isopropanol, just the time needed for coating all the samples, as the SAM may be released.)
- Dry just **the sample of interest** with the Nitrogen gun right before BAMS. Leave the others in immersion.
- Waste: throw away the solutions in NO HALOGENATED.
- COAT THE SAMPLES RIGHT AFTER SAM!!!!

7. BAMS

- Switch on the hotplate of the BAMS at 105°C
- Once the temperature is achieved, leave the semiconductor vials onto the hotplate along the coating process. The semiconductor must be always hot.
- Bar number: **3, 4 or 5**. Clean the bar with ACETONE and then ISOPROPANOL with a wipe. Dry with Nitrogen gun.

- Leave the bar already placed at least 20-30 min before coating. (the bar must be hot)
- The height of a glass slide is the proper distance between the bar and the hotplate. Check with the glass slide a homogenous distance along the bar. In case of doubt, please contact with your colleagues.
- Fix the sample onto the hotplate by introducing Chlorobenzene between the hotplate and the sample (use a glass paster pipette)
- Ink amount around 30-40 μ L for each sample (for a 1.5-2 cm x 2.5-3 cm). **The less, the better.** Attention: avoid bubbles by placing the solution in the upper or central part of the bar.
- Guide the channels of the OFETs to be the most parallel/perpendicular to the bar.
- Proceed to coat the samples **VERY QUICKLY once the meniscus is created (otherwise the solution will start to evaporate)**. Recommendation: one hand to place the solution into the bar and the other hand to press the button Jog + at the PC.
- Speed: 1-10 mm/s. Acceleration: 0 mm/s. NOTE: the equipment allows speeds up to 100 mm/s. However, it is not recommended to overcome 50 mm/s
- Remove **quickly** the sample from the hotplate to avoid an extra curing of the semiconductor.

SAMPLES DONE!